Exact Analytic Solution of a Boundary Value Problem for the Falkner–Skan Equation

By P. L. Sachdev, Ramesh B. Kudenatti, and N. M. Bujurke

The Falkner–Skan equation, subject to appropriate physical boundary conditions arising from boundary layer theory, is exactly solved. The results obtained from this solution are compared with the numerical solution. The Blasius equation, subject to the same boundary conditions, is also solved exactly; the solution is compared with the earlier work on this equation. The analytic solution presented here agrees closely with the corresponding numerical results.

1. Introduction

One of the most celebrated nonlinear ODEs which appear in fluid mechanics is referred to as Falkner–Skan equation. It is solved over $(0, \infty)$ subject to the boundary conditions at 0 and ∞ arising from the flow conditions. The Falkner–Skan equation arises from the similarity reduction of nonlinear PDEs describing flow in the boundary layer on a flat plate which has a constant velocity in the direction opposite to that of a uniform mainstream (see Section 2 for a brief derivation of (6)–(7) from the basic system). Existence and uniqueness of the solution of (6)–(7) was first discussed by Weyl [1] for $\lambda = 0$. Hussaini et al. [2] showed that the solution of (6)–(7) was non-unique for $\lambda > 0$; they also gave an upper bound for the critical value of $\lambda = \lambda_c$

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beyond which no solution exists. Coppel [3] proved the results of Weyl [1] in a constructive manner. Specifically, he proved the following existence and uniqueness theorem: For all non-negative $f(0)$ and $f'(0)$, the second derivative f'' is positive, zero or negative throughout the interval $0 \leq \eta \leq \infty$ according as $f'(0)$ is less than, equal to or greater than 1. With this restriction on f'', the solution of the problem is unique. He also investigated in some detail the qualitative properties of the solution of (6)–(7). Coppel [4] solved (6) exactly for $\beta = -1$ in terms of parabolic cylinder functions.

Riley and Weidman [5] undertook a detailed numerical study of the boundary value problem (6)–(7). They discovered a rich structure of the solution of (6)–(7) for $|\beta| \leq 1$ over a range of values of λ, both positive and negative. In particular, their numerical solution showed that, for $-1 \leq \beta \leq 0$, the system (6)–(7) admits dual solutions below a critical value of the parameter $\lambda = \lambda_m(\beta)$, say, and no solution if $\lambda > \lambda_m(\beta)$. Further, for $\lambda > 0$, they found a unique solution for $0.14 \leq \beta \leq 0.5$, dual solutions for $0.5 < \beta \leq 1$ and even triple solutions for $0 < \beta \lesssim 0.14$. Afzal [6] recently considered another version of the boundary value problem for the Falkner–Skan equation. He solved the problem exactly for the special case $\beta = -1$. He also found asymptotic form of the solution both when β is small and when it is large.

In the present paper, we give an exact analytic solution of (6)–(7) in the form

$$f(\eta) = \eta + \delta - \frac{\delta}{G(\eta)}, \tag{1}$$

where the function $G(\eta)$ is defined by an infinite power series in η. This form is inspired by the single hump solution (12) of Burgers equation (see also Whitham [7], Sachdev [8]). This form has a close resemblance with the exact solution of (6)–(7) for $\beta = -1$ (cf. (9)–(12)). That such resemblance should occur is not entirely surprising since nonlinear convection and viscosity are common features of both the boundary layer equations and the Burgers equation. One must also observe the differences in the BVP (6)–(7) and the BVP for the ODE resulting from the Burgers equation and the corresponding boundary conditions. The former is a third order nonlinear ODE with non-zero f' as $\eta \to \infty$ while the latter is a second order nonlinear ODE with $f \to 0$ as $\eta \to \pm\infty$. The reciprocal function $H = \frac{1}{f(\eta)}$ enabled Sachdev and his collaborators (Sachdev et al. [9, 10], Sachdev and Nair [11], Sachdev and Srinivasa Rao [12]) to discover a new class of ODEs, referred to as Euler–Painlevé transcendents, which unify the study of generalized Burgers equations just as Painlevé transcendents characterize K-dV type of equations; their analytic properties are, however, quite different.

In Section 3, we briefly derive the exact solution of (6)–(7) for $\beta = -1$ and hence write out the solution form of this system for general β. This leads to an exact solution of (6)–(7) involving two parameters. One of these constants is found by making the solution (14) and (18) of (6)–(7) tend to the special solution (9) as $\beta \to -1$. The other constant is determined by finding $f''(0)$ from an integral relation arising from (6) which satisfies all the conditions in (7). The iterative solution of this integral relation for $f''(0)$ leads to one, double or no solution depending on the parameters λ and β. The results obtained from our exact solution agree remarkably with the numerical solution of Riley and Weidman [5].

The present paper is organized as follows. In Section 2, a brief derivation of the Falkner–Skan equation with appropriate boundary conditions from the basic fluid dynamic system is given; its exact solution for $\beta = -1$ is also explicated. Section 3 is devoted to exact analytical solution of the problem (6)–(7) for general β. In Section 4, we deal with the asymptotic behavior of the Falkner–Skan equation as $\eta \to 0$ and $\eta \to \infty$. In Section 5, we give a detailed analysis of the Blasius equation- its exact solution subject to relevant boundary conditions as well as comparison with the earlier analysis of this equation (Meksyn [13]). In Section 6, we compare the results obtained by our exact and asymptotic solutions with the numerical results of Riley and Weidman [5]. The concluding section summarizes the major results of the present study.

2. Formulation of the problem

Boundary layer flows on continuous stretching surfaces, moving in an otherwise quiescent fluid medium, have received considerable attention due to their applications in industrial processes. Examples are aerodynamic extrusion of plastic sheets, boundary layer flows along liquid films in condensation processes, cooling of a metallic plate in a cooling bath etc. One can assume that stretching varies with distance from the die according to a power law. Also, as the ambient fluid is stationary far away from the surface, the flow is governed by boundary layer equations. Similarity solutions for stretching sheet problems have been studied by Crane [14], Banks and Zaturska [15], Magyari and Keller [16], and others, referred to therein.

The two-dimensional laminar boundary layer equations of an incompressible fluid subjected to a pressure gradient are

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{1}{\rho}p' + \nu\frac{\partial^2 u}{\partial y^2}, \tag{2}$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0, \tag{3}$$

where p' is the pressure gradient, $p' = -\rho U\frac{dU}{dx}$, u is the streamwise velocity in the direction of the fluid flow, v is the velocity in the direction normal to u, ν is the constant kinematic viscosity, $U(x)$ is the velocity at the edge of the boundary layer which obeys the power-law relation $U(x) = U_\infty x^m$. The relevant boundary conditions are

$$y = 0: \quad u = U_w(x), \quad v = 0; \quad u \to U_\infty \quad \text{as} \quad y \to \infty, \tag{4}$$

where U_w is the stretching surface velocity which obeys the power law relation, $U_w = U_{0w} x^m$. Using the similarity transformation

$$\psi = \sqrt{\frac{2\nu x U(x)}{1+m}} f(\eta), \quad \eta = y\sqrt{\frac{(1+m)U(x)}{2\nu x}}, \quad \beta = \frac{2m}{1+m} \tag{5}$$

in Equations (2)–(4) and simplifying, we get a third order nonlinear ODE (see Yuan [17] for details)

$$f''' + ff'' + \beta(1 - f'^2) = 0, \quad ' = \frac{d}{d\eta} \tag{6}$$

with the boundary conditions

$$f(0) = 0, \quad f'(0) = -\lambda, \quad f'(\infty) = 1, \tag{7}$$

where $\lambda = -\frac{U_w}{U_\infty}$. For $\beta \geq 0$, Equation (6) describes the symmetrical boundary layer flow over a wedge of included angle $\beta\pi$. For $\lambda \neq 0$, the boundary has a precise speed and stretch.

Integrating (6) with $\beta = -1$ twice with η, we get a Riccati type equation

$$f' + \frac{f^2}{2} = \frac{\eta^2}{2} + \delta\eta - \lambda, \tag{8}$$

where $\delta = f''(0)$. The solution of (8) subject to (7) may be found to be

$$f(\eta) = \eta + \delta - \frac{\delta \exp\left(-\left(\delta\eta + \frac{\eta^2}{2}\right)\right)}{1 - \sqrt{\frac{\pi}{2}}\frac{\delta}{2}\exp\left(\frac{\delta^2}{2}\right)\left(\operatorname{erf}\left(\frac{\eta+\delta}{\sqrt{2}}\right) - \operatorname{erf}\left(\frac{\delta}{\sqrt{2}}\right)\right)} \tag{9}$$

provided that $\delta^2 = -2(1 + \lambda)$.

The axial velocity gradient at the wall is obtained from (8) as

$$f''(0) = \delta = \pm\sqrt{-2(1+\lambda)}. \tag{10}$$

The velocity profile $f'(\eta)$ may be easily obtained from (9). Equation (10) shows that, for $\beta = -1$, there are dual solutions for $\lambda < -1$ and no solution for $\lambda > -1$. It may also be observed that for $\lambda = -1$, $f(\eta) = \eta$ is an exact solution of (6)–(7).

3. Exact solution of the Falkner–Skan equation

Inspired by the first author's work on generalized Burgers equations and their reduction to Euler–Painlevé transcedents to unify their character with reference to the single hump solution, we seek exact solution of (6)–(7). The similarity solution of the Burgers equation

$$u_t + uu_x = \frac{\delta}{2} u_{xx}, \tag{11}$$

for the so-called single hump solution is

$$u = \left(\frac{\delta}{t}\right)^{1/2} \frac{\exp(-\xi^2)}{\dfrac{\sqrt{2\pi}}{\exp(R) - 1} + \sqrt{\dfrac{\pi}{2}}\mathrm{erf}(\xi)}, \tag{12}$$

where $\xi = \frac{x}{\sqrt{2\delta t}}$ and R is the Reynolds number, that is, the ratio of area under the profile divided by δ; this solution vanishes as $\xi \to \pm\infty$. It is striking that the form (9) of the solution for the Falkner–Skan system (6)–(7) for $\beta = -1$ and the similarity solution (12) of the Burgers equation (11) are so close. However, the solution $f(\eta)$ of the Falkner–Skan equation (6), which is of third order, satisfies a quite different condition, namely, $f'(\eta) \to 1$ as $\eta \to \infty$, hence the term $\eta + \delta$ in (9).

We rewrite the solution (9) for $\beta = -1$ as

$$\begin{aligned} f &= \eta + \delta \\ &- \frac{\delta}{\left(\exp(-\delta^2/2) + \sqrt{\frac{\pi}{2}}\frac{\delta}{2}\mathrm{erf}\left(\frac{\delta}{\sqrt{2}}\right)\right)\exp((\eta+\delta)^2/2) - \sqrt{\frac{\pi}{2}}\frac{\delta}{2}\exp((\eta+\delta)^2/2)\mathrm{erf}\left(\frac{\eta+\delta}{\sqrt{2}}\right)}. \end{aligned} \tag{13}$$

Motivated by the form (13) which is solution of (6)–(7) for $\beta = -1$, we let

$$f(\eta) = \eta + \delta - \frac{\delta}{G(\eta)} \tag{14}$$

for general β. Substituting (14) into (6) we get the following equation for $G(\eta)$:

$$\begin{aligned} &G^2G''' - G[6G' + \delta - (\eta + \delta)G]G'' + 6G'^3 \\ &\quad + [\delta(2 - \beta) - 2(\eta + \delta)G]G'^2 - 2\beta G^2G' = 0. \end{aligned} \tag{15}$$

The boundary conditions (7) now become

$$G(0) = 1, \quad G'(0) = \frac{\delta}{2}, \quad G(\infty) = \infty. \tag{16}$$

The solution of (15) for $\beta = -1$, subject to (16), is given by

$$G(\eta) = \left(\exp(-\delta^2/2) + \sqrt{\frac{\pi}{2}}\frac{\delta}{2}\operatorname{erf}\left(\frac{\delta}{\sqrt{2}}\right)\right)\exp((\eta+\delta)^2/2)$$
$$-\sqrt{\frac{\pi}{2}}\frac{\delta}{2}\exp((\eta+\delta)^2/2)\operatorname{erf}\left(\frac{\eta+\delta}{\sqrt{2}}\right). \qquad (17)$$

Motivated by the series representation of the solution (17) for $\beta = -1$, we assume

$$G = \sum_{n=0}^{\infty} a_n \eta^n, \qquad (18)$$

for general β. Substituting (18) into (15) and equating the coefficients of η^n to zero, etc., we get

$$a_0 = 1, a_1 = \frac{\delta}{2}, a_3 = \frac{\delta}{24}\left(-3\delta^2 + \beta(4+\delta^2) + 24a_2\right),$$
$$a_4 = \frac{1}{96}\left(\delta^2(2 - 5\delta^2 + 2\beta(6+\delta^2)) + 4(-2 + 5\delta^2 + 2\beta(2+\delta^2))\,a_2 + 96a_2^2\right),$$
$$a_5 = \frac{\delta}{480}\left(2\delta^2(5+2\delta^2) + \beta^2(8 + 6\delta^2 + \delta^4) - 2\beta(4 + 3\delta^2 + 2\delta^4)\right.$$
$$\left. + 4(-10 - 33\delta^2 + 4\beta(15 + 4\delta^2))\,a_2 + 32(22+\beta)\,a_2^2\right),$$

and, in general,

$$a_{n+3} = -\frac{1}{(n+1)(n+2)(n+3)}\left[\sum_{m=0}^{n-1}\sum_{k=0}^{n-m}(m+1)(m+2)(m+3)a_k a_{n-m-k} a_{m+3}\right.$$
$$+ \sum_{m=0}^{n}(-\delta(m+1)[(m+2)\,a_{n-m}a_{m+2} - (2-\beta)(n-m+1)a_{m+1}a_{n-m+1}]$$
$$- \sum_{k=0}^{m}(k+1)(m-k+1)[2\delta a_{n-m} - 6(n-m+1)a_{n-m+1}]\,a_{k+1}a_{m-k+1}$$
$$+ \sum_{k=0}^{n-m}((m+1)[m a_{m+1} + \delta(m+2)a_{m+2} - 2\beta a_{m+1}]a_k a_{n-m-k} - (k+1)$$
$$\left.\times[6(m+1)(m+2)\,a_{m+2} + 2(n-m-k)\,a_m]a_{k+1}a_{n-m-k}))\right],$$
$$n = 1, 2, ---, \qquad (19)$$

where the coefficients a_n have been expressed in terms of a_2 which itself related to $f''(0)$ by

$$a_2 = \left(f''(0) + \frac{\delta^3}{2}\right) \bigg/ 2\delta. \tag{20}$$

The coefficients a_n involve two arbitrary constants, namely, δ and $f''(0)$. We match the series (18) with the series form of the exact solution (17) for $\beta = -1$. This yields $\delta = \sqrt{-2(1+\lambda)}$. The other constant a_2 or $f''(0)$ remains to be determined. Thus, we have infinite solutions of (6)–(7) in the form (14). We determine the constant $f''(0)$ in the following manner. Integrating (6) from $\eta = 0$ to $\eta = \infty$ and using (7), we get

$$\int_0^\infty (f' - f'^2)\, d\eta + \beta \int_0^\infty (1 - f'^2)\, d\eta = f''(0). \tag{21}$$

$f''(0)$ also appears on the LHS of (21) in the series form of f', obtained from the solution (14) and (18). We solve (21) iteratively to obtain the numerical value $f''(0)$ for different sets of parameters β and λ. $f''(0)$ found from (21) agrees very well with that obtained by solving the BVP (6)–(7) numerically (see Tables 1 and 2).

4. Asymptotic solutions

We have found asymptotic forms of the solution of the BVP (6)–(7) as $\eta \to 0$ and $\eta \to \infty$. We write Taylor series form of the solution

$$f = \sum_{n=0}^{\infty} b_n \eta^n, \tag{22}$$

of Equation (6) as $\eta \to 0$ and, using the first two conditions in (7), obtain

$$\begin{aligned}
&b_0 = 0,\;\; b_1 = -\lambda,\;\; b_3 = -\frac{1}{6}\beta(1-\lambda^2),\;\; b_4 = \frac{1}{12}\lambda(1-2\beta)\, b_2,\\
&b_5 = \frac{1}{60}\big(-\beta\lambda(-1+\beta)(-1+\lambda^2) + (-2+4\beta)b_2^2\big),\\
&b_{n+3} = -\frac{1}{(n+1)(n+2)(n+3)}\left(\sum_{m=0}^{n}(m+1)(m+2)\, b_{n-m} b_{m+2}\right.\\
&\qquad\left. -\beta \sum_{m=0}^{n}(m+1)(n-m+1)\, b_{m+1} b_{n-m+1}\right), \quad n = 1, 2, ---,
\end{aligned} \tag{23}$$

where b_2 is an arbitrary constant. We seek solution at the other end in the form $f = \eta + \eta_0 + E(\eta)$, $|E'(\eta)| \ll 1$, which identically satisfies the condition $f'(\infty) = 1$. Substituting this form in (6), we get

$$E''' + \xi E'' - 2\beta E' = 0, \tag{24}$$

$$E(0) = -\eta_0, \quad E'(0) = -(1+\lambda), \quad E'(\infty) = 0, \tag{24$_1$}$$

where $\xi = \eta + \eta_0$. Choosing the solution of (24) which tends to zero as $\eta \to \infty$, we have

Table 1
Comparison of the Exact and the Numerical Solution of (6)–(7) for $\lambda = -1.4$

η	$f'(\eta)$ for $\beta = 1.0$ Exact Solution	$f'(\eta)$ for $\beta = 1.0$ Numerical Solution	$f'(\eta)$ for $\beta = 2.0$ Exact Solution	$f'(\eta)$ for $\beta = 2.0$ Numerical Solution
0.0	1.4	1.4	1.4	1.4
0.125	1.321262788060	1.32126326590	1.29987151975	1.2998763568
0.250	1.255713730163	1.25571444328	1.22325522164	1.2232622903
0.375	1.201649877542	1.20165060445	1.16495846700	1.1649660000
0.500	1.157491693875	1.15749223272	1.12089622178	1.1209030048
0.625	1.121788593434	1.12178876070	1.08784356306	1.0878487466
0.750	1.093223618112	1.09322324777	1.06325417442	1.0632571466
0.875	1.070616151717	1.07061509447	1.04512290902	1.0451232069
1.000	1.052922003421	1.05292012600	1.03187863648	1.0318758827
1.125	1.039230600870	1.03922778530	1.02229870302	1.0222925592
1.250	1.028759381381	1.02875552419	1.01543951291	1.0154296454
1.375	1.020845735208	1.02084074680	1.01057968106	1.0105657345
1.500	1.014937029641	1.01493083496	1.00717336602	1.0071549426
1.625	1.010579329032	1.01057187310	1.00481207055	1.0047887124
1.750	1.007405433322	1.00739670249	1.00319358835	1.0031647615
1.875	1.005122799437	1.00511288670	1.00209700641	1.0020620821
2.000	1.003501799471	1.00349108142	1.00136281874	1.0013210495
2.125	1.002364620402	1.00235419015	1.00087731470	1.0008277994
2.250	1.001574945027	1.00156756814	1.00056049633	1.0005021289
2.375	1.001028434957	1.00103042467	1.00035686640	1.0002882611
2.500	1.000644106435	1.00066847485	1.00022851338	1.0001478999
2.625	1.000357119388	1.00042781392	1.00015000372	1.0000550857
2.750	1.000114107514	1.00026992948	1.00010467099	1.0000151721
2.875	1.000090250771	1.00016773087	1.00006015625	1.0000091492
3.000	1.000051862143	1.00010245582	1.00004105762	1.0000060858
3.125	1.000036025407	1.00006131056	1.00002925031	1.0000038612
3.250	1.000006166911	1.00003570573	1.00002167103	1.0000001857

$$f = \eta + \eta_0 + 2^{-1-2\beta}\pi C_1 \left(-\frac{\sqrt{\frac{2}{\pi}}\left(-1 + M\left(\hat{a}_1, \hat{b}_1, \frac{-(\eta+\eta_0)^2}{2}\right)\right)}{(1+2\beta)\Gamma\left(\frac{1+2\beta}{2}\right)} + \frac{(\eta+\eta_0) M\left(\hat{a}_2, \hat{b}_2, \frac{-(\eta+\eta_0)^2}{2}\right)}{\sqrt{\pi}\,\Gamma(1+\beta)} \right), \tag{25}$$

Table 2
Comparison of the Exact Solution of (26)–(27) with that of Meksyn (30) and Numerical Solution for $\lambda = -1.4$

η	Exact solution $f'(\eta)$	Meksyn's approach $f'(\eta)$	Numerical solution $f'(\eta)$
0.0	1.4	1.4	1.4
0.125	1.355493528671741	1.355560808966235	1.35549361017901
0.25	1.311930745138507	1.311963905707501	1.31193088389411
0.375	1.270169576972004	1.270165983634023	1.27016974767493
O.5	1.230933798667421	1.230939812107345	1.23093398101678
0.625	1.194782985937845	1.194784217214935	1.19478316903746
0.75	1.162099022813424	1.162099589695037	1.16209920694593
O.875	1.133087877943110	1.133084642362781	1.13308807358128
1.0	1.107793898925250	1.107793672844840	1.10779412356508
1.125	1.086123081543761	1.086124795297727	1.08612335535609
1.25	1.067871594488475	1.067878342782991	1.06787193588310
1.375	1.052756138518348	1.052759362033306	1.05275656060372
1.5	1.040443322792912	1.040442324114990	1.04044383159121
1.625	1.030575990872846	1.030575153154484	1.03057658627030
1.75	1.022795196444082	1.022793875190888	1.02279587751753
1.875	1.016757224292779	1.016762573551040	1.01675800213791
2.0	1.012145622157324	1.012146291699510	1.01214654583167
2.125	1.008678630813461	1.008673082507466	1.00867983544244
2.25	1.006112672807225	1.006127043481449	1.00611446289276
2.375	1.004242699751609	1.004230707770536	1.00424568451432
2.5	1.002900228288396	1.002859368928471	1.00290552844538
2.625	1.001949843911165	1.001826864572927	1.00195938958383
2.75	1.001284848051877	1.001120849974007	1.00130178470331
2.875	1.000822592442239	1.000714383786236	1.00085180556738
3.0	1.000499906344290	1.000500591982737	1.00054866644573
3.125	1.000268891852083	1.000216623215812	1.00034761003747
3.25	1.000093249667934	1.000084152673725	1.00021632246421
3.375	1.000010875462154	1.000016870868170	1.00013191862257

where $\hat{a}_1 = -\frac{1}{2} - \beta$, $\hat{b}_1 = \frac{1}{2}$, $\hat{a}_2 = -\beta$, $\hat{b}_2 = \frac{3}{2}$ and $M(\hat{a}, \hat{b}, \eta)$ is the confluent hypergoemetric function; η_0 and C_1 are arbitrary constants.

The unknown coefficients b_2, η_0 and C_1 in (22) and (25) were found numerically. The agreement of these asymptotic solutions with the numerical results is rather remarkable. The asymptotic forms (22) and (25) hold for all values of β and λ.

5. Boundary value problem for the Blasius equation

With $\beta = 0$, Equation (6) reduces to the so-called Blasius equation. The boundary value problem (6)–(7) now becomes

$$f''' + ff'' = 0, \tag{26}$$

$$f(0) = 0, \quad f'(0) = -\lambda, \quad f'(\infty) = 1, \tag{27}$$

and describes the boundary layer flow on a continuous flat plate in which both the free stream and plate velocities are constant with the same sign.

Substituting (14) into (26), we get

$$G^2G''' - G\left[6G' + \delta - (\eta+\delta)G\right]G'' + 6G'^3 + 2\left[\delta - (\eta+\delta)G\right]G'^2 = 0, \tag{28}$$

which must be solved subject to the conditions (16). First few coefficients of the solution (18) of (28) subject to the first two conditions in (16) are

$$a_0 = 1, a_1 = \frac{\delta}{2}, a_3 = \frac{-\delta^3}{8} + \delta a_2, a_4 = \frac{\delta^2}{96}(2 - 5\delta^2) + \frac{1}{24}(-2 + 5\delta^2)\, a_2 + a_2^2,$$

$$\begin{aligned} a_{n+3} = -\frac{1}{(n+1)(n+2)(n+3)}\Bigg[&\sum_{m=0}^{n-1}\sum_{k=0}^{n-m}(m+1)(m+2)(m+3)a_k a_{n-m-k} a_{m+3} \\ &+ \sum_{m=0}^{n}(-\delta(m+1)[(m+2)\, a_{n-m}a_{m+2} - 2(n-m+1)a_{m+1}a_{n-m+1}] \\ &- \sum_{k=0}^{m}(k+1)(m-k+1)[2\delta a_{n-m} - 6(n-m+1)a_{n-m+1}]\, a_{k+1}a_{m-k+1} \\ &+ \sum_{k=0}^{n-m}((m+1)[m a_{m+1} + \delta(m+2)\, a_{m+2}]a_k a_{n-m-k} - (k+1) \\ &\times [6(m+1)(m+2)a_{m+2} + 2(n-m-k)a_m]a_{k+1}a_{n-m-k}))\Bigg], \end{aligned}$$

$$n = 1, 2, ---. \tag{29}$$

Here, again, $\delta = \sqrt{-2(1+\lambda)}$; a_2 remains unknown and must be obtained iteratively from (21) with $\beta = 0$.

It is instructive to compare our exact analytic solution of the Blasius equation (26) subject to (27) with that obtained earlier by Meksyn [13] for the special case $\lambda = 0$. Substituting the power series form of the solution

$$f = \sum_{m=0}^{\infty} \frac{c_m}{m!}\eta^m \tag{30}$$

into (26), we get

$$c_0 = 0, \quad c_1 = -\lambda, \quad c_4 = \lambda c_2, \quad c_5 = -c_2^2, \quad c_6 = 3\lambda^2 c_2,$$
$$c_7 = -11\lambda c_2^2, \quad c_8 = 15\lambda^3 c_2 + 11c_2^2 \text{ etc.,} \tag{31}$$

where the constant $c_2 = f''(0)$ is arbitrary. Integrating Equation (26) twice and using the conditions (27), we get

$$1 + \lambda = c_2 \int_0^\infty \exp(-F(\eta))\, d\eta, \tag{32}$$

where

$$F(\eta) = \int_0^\eta f(\eta)\, d\eta. \tag{33}$$

To determine the constant c_2, Meksyn [13] introduced a new variable τ:

$$F(\eta) = \int_0^\eta f(\eta)\, d\eta \equiv \tau. \tag{34}$$

Substituting (30) into (34) and inverting the series, one may obtain

$$\eta = \sqrt{-\frac{2}{\lambda}}\sqrt{\tau} - \frac{c_2}{3\lambda^2}\tau - \frac{\sqrt{-\frac{1}{\lambda}}\left(5c_2^2 + 3\lambda c_3\right)}{18\sqrt{2}\lambda^3}\tau^{3/2}$$
$$+ \frac{40c_2^3 + 45\lambda c_2 c_3 + 9\lambda^2 c_4}{270\lambda^5}\tau^2 + - - -. \tag{35}$$

Now using (34) and (35) in (32), we have

$$1 + \lambda = c_2 \int_0^\infty \exp(-\tau)\frac{d\eta}{d\tau}d\tau = c_2 \sum_{m=0}^\infty d_m \Gamma\left(\frac{m+1}{2}\right), \tag{36}$$

where

$$d_0 = \frac{1}{\sqrt{2}}\sqrt{-\frac{1}{\lambda}}, d_1 = -\frac{c_2}{3\lambda^2}, d_2 = \frac{5\left(-\frac{1}{\lambda}\right)^{7/2} c_2^2}{12\sqrt{2}}, d_3 = \frac{9\lambda^3 c_2 + 40c_2^3}{135\lambda^5} \text{ etc.} \tag{37}$$

and Γ represents the Gamma function. Thus, we may find c_2 explicitly from the Equation (36). Meksyn [13] affirmed that the series (30) converges rapidly. $f''(0)$ for different values of λ as obtained by Meksyn's approach as well as from (21) with $\beta = 0$ agree very well with that found numerically (see Table 2).

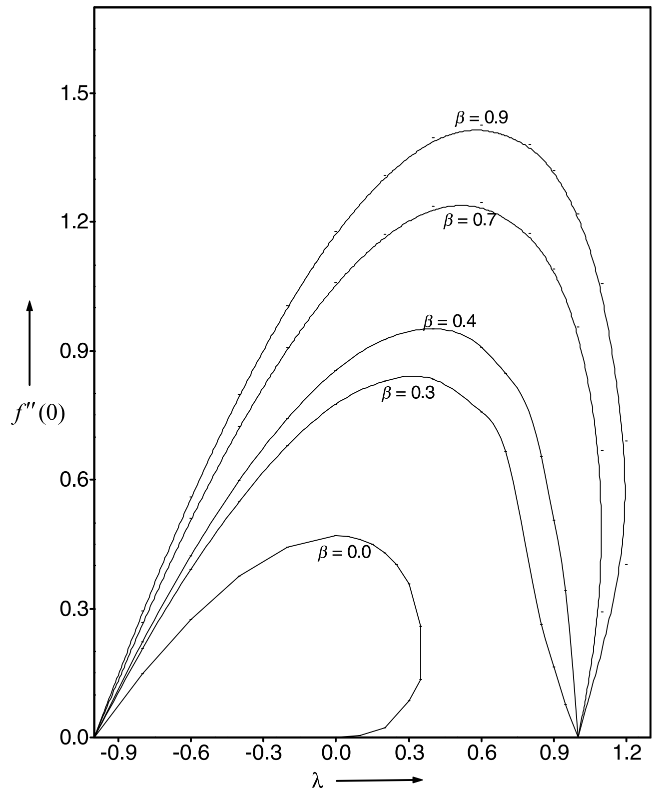

Figure 1. Velocity gradient at the wall $f''(0)$ versus λ for different values of parameter β.

6. Results and discussions

The exact analytic solution (14) and (18) of the Falkner–Skan equation (6), subject to the boundary conditions (7), which simulates and embeds the known exact solution (9) for $\beta = -1$ gives a remarkable insight into the problem. This form, as it is, gives infinite solutions of (6) which satisfy the conditions (7): it involves one arbitrary parameter a_2 or $f''(0)$ (see Equation (20)). This parameter must be found from the integral relation (21) which itself is derived from (6)–(7). This relation, however, is highly implicit. Thus, equation (21) must be solved numerically for a_2 in a recursive manner. The results thus obtained for different values of the parameter β are compared with the direct numerical solution of (6)–(7) and are shown in Table 1. We use Pade' approximants for summing the series (18) for G. $f''(0)$ is plotted against λ in Figure 1 for different values of β. Our results closely agree with those of Riley and Weidman [5] obtained numerically. For $\beta = 0$, we get dual solutions for $0 < \lambda < 0.3546$ and no solution for $\lambda > 0.3546$. For $0.5 < \beta < 1.0$, there are dual solutions for $\lambda > 1$ while there is

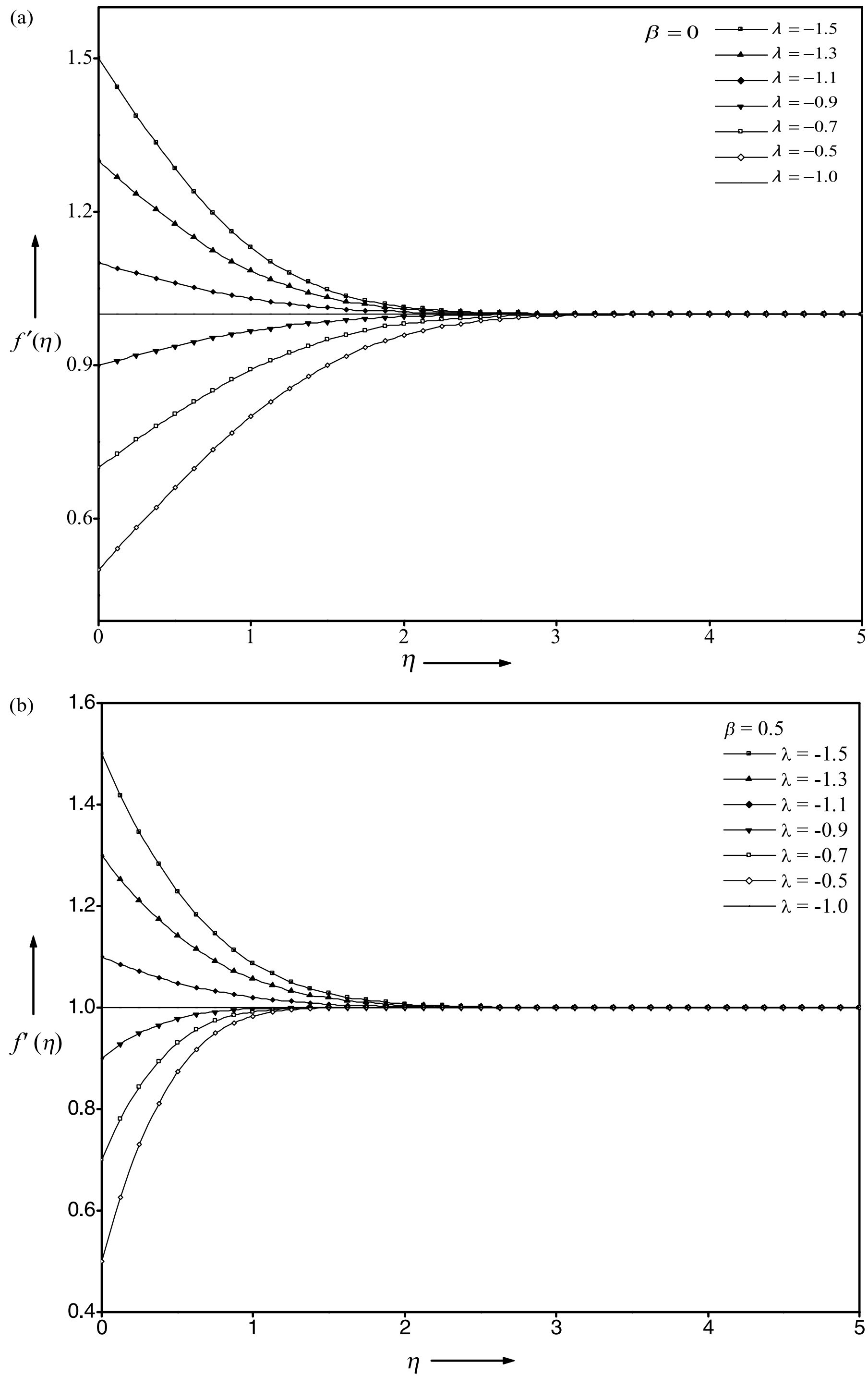

Figure 2. Velocity distribution $f'(\eta)$ against η for different stretching parameter β.

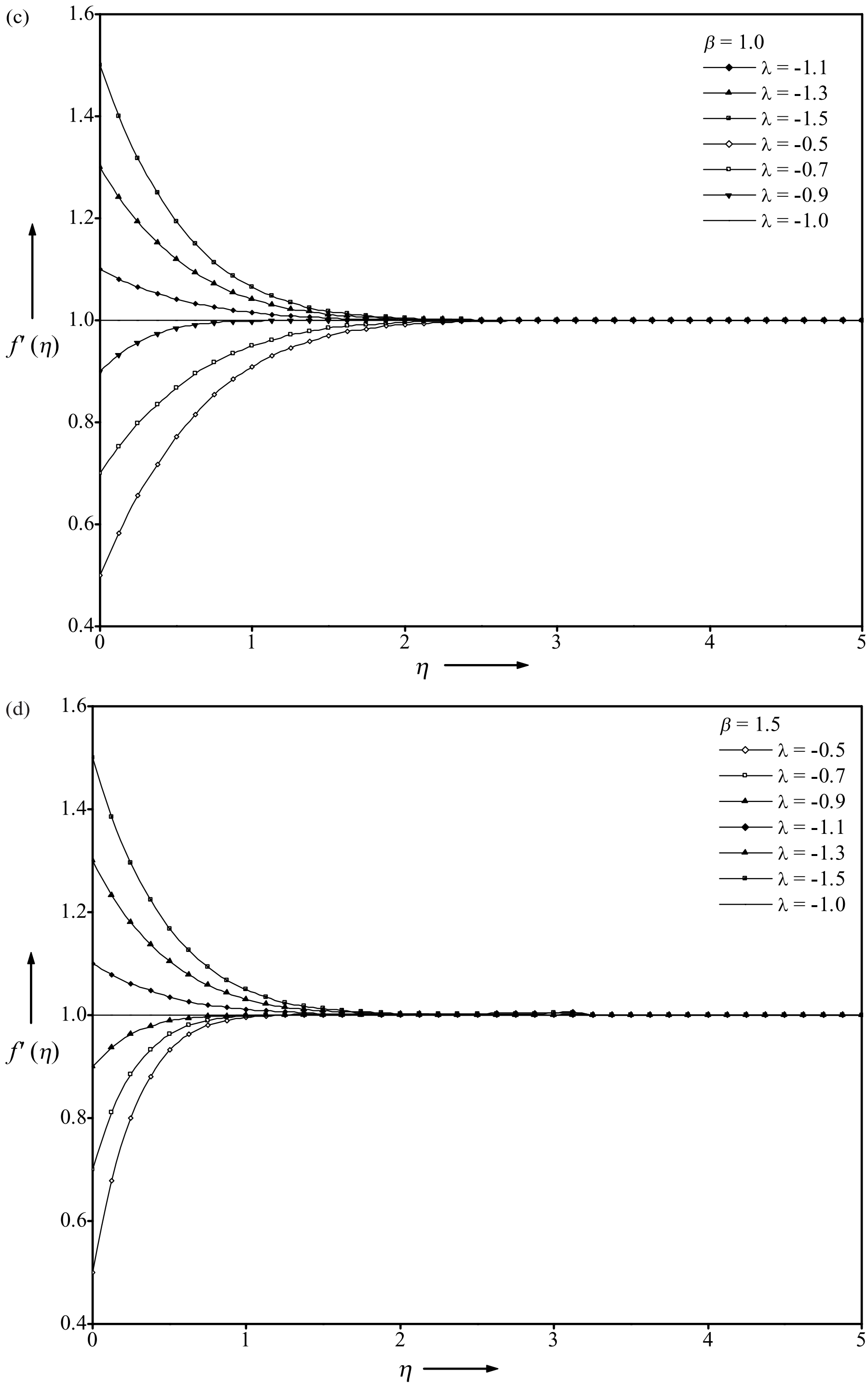

Figure 2. (*continued*)

only one solution for $\lambda < 1$. For $0.14 < \beta < 0.5$, there is a unique solution for all $\lambda > 0$. However, we have been unable to recover the triple solution for $0 < \beta < 0.14$, claimed by Riley and Weidman [5]. Figures 2(a–d) show $f'(\eta)$ versus η for different sets of the parameters λ and β. The approach of the solution to its asymptotic value 1 as $\eta \to \infty$ is clearly seen. We may observe from Figure 2 that the special exact solution $f = \eta$ of (6)–(7) with $\lambda = -1$ demarcates qualitatively different behavior of the solutions for $\lambda > -1$ and $\lambda < -1$.

Table 2 compares the solution for the Blasius equation (26) subject to (27) obtained by our exact method, that of Meksyn [13] explained in Section 5 and the numerical results of Riley and Wiedman [5]; a close agreement of the results from the three approaches is clearly observed. Meksyn's series expansion gives good results even with a few terms. We may also observe that the Taylor series solution found in Section (4) when supplemented by the value $f''(0)$ obtained numerically gives good accuracy even for large η.

7. Conclusions

Motivated by the close analytical resemblance of the exact solution (9) of Falkner–Skan equation (6) with $\beta = -1$, subject to (7), and the so-called single hump solution (12) of Burgers equation (Sachdev [8]), we have found an exact analytic solution of the former for general β. This new solution satisfies the boundary conditions (7) and embeds the special exact solution (9) for $\beta = -1$. The exact solution (14) and (18) of (6)–(7) together with the integral relation (21) confirms all the qualitative features of the solution observed earlier numerically by Riley and Weidman [5]. We hope to extend the present analysis to Falkner–Skan equation and its kindred class with the general boundary conditions $f(0) = \alpha$, $f'(0) = \gamma$ and $f'(\infty) = 1$, where α and γ are real.

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